



Ilia Iliashenko

Contacts

iliailiashenko@gmail.com

Location

Antwerp, Belgium

Citizenship

Belgium, Russia

Interests

Privacy enhancing technologies, secure computation methods, fully homomorphic encryption, secure multi-party computation, zero-knowledge proofs

Contributions

I contributed to several software libraries implementing privacy-enhancing technologies. In particular,

- [World ID biometric authentication \(Proof of identity via iris matching\)](#), Rust
- [Ciphercore \(ABY3 secure multi-party computation protocol\)](#), Rust/Python,
- [FINAL \(FINAL fully homomorphic encryption scheme\)](#), C++),
- [Microsoft SEAL \(CKKS/HEAAN homomorphic encryption scheme\)](#), C++).

Skills

Human languages	Russian (native) English (full proficiency) Dutch (intermediate) German (elementary)
Programming languages	Rust, C++, Python, Protobuf SageMath, Magma, R LaTeX, Circom
IDEs	VS Code, Sublime Text, Zed
Agentic AI	Claude Code

Employment

Oct. 2024 – Jun. 2026	Research engineer Inversed Tech, Remote
Aug. 2021 – Sep. 2024	Research engineer Ciphermode Labs DBA Pyte, Remote
May 2019 – Aug. 2021	Postdoctoral researcher ESAT/COSIC, KU Leuven, Leuven, Belgium
Jun. 2019 – Sep. 2019, Jun. 2018 – Sep. 2018	Research intern Cryptography and Privacy Research group, Microsoft Research, Redmond, WA, USA
Sep. 2013 – Aug. 2015	Postgraduate researcher IKBFU, Kaliningrad, Russia
Aug. 2012 – Jun. 2014	C++ programmer Mariaglorum, Kaliningrad, Russia

Education

Aug. 2015 – May 2019

Ph.D. in Engineering Science
“Optimisations of fully homomorphic encryption”
Supervisors: Prof. Bart Preneel,
Prof. Frederik Vercauteren
ESAT, KU Leuven, Belgium

Sep. 2007 – Jan. 2013

Diploma in Mathematics (summa cum laude)
“Quantum security of the McEliece public-key cryptosystem”
Supervisor: Dr. Sergey Aleshnikov
IKBFU, Kaliningrad, Russia

Publications

R. Geelen, I. Iliashenko, Jiayi Kang and F. Vercauteren,
On Polynomial Functions Modulo p^e and Faster Bootstrapping for Homomorphic Encryption,
Advances in Cryptology - EUROCRYPT 2023 - 42nd Annual International Conference on the Theory and Applications of Cryptographic Techniques, Lyon, France, April 23-27, 2023, Proceedings, Part III, volume 14006 of Lecture Notes in Computer Science, C. Hazay and M. Stam (eds.), pp. 257-286, Springer-Verlag, 2023

C. Bonte, I. Iliashenko, J. Park, H. V. L. Pereira and N. P. Smart,
FINAL: Faster FHE Instantiated with NTRU and LWE,
Advances in Cryptology - ASIACRYPT 2022 - 28th International Conference on the Theory and Application of Cryptology and Information Security, Taipei, Taiwan, December 5-9, 2022, Proceedings, Part II, volume 13792 of Lecture Notes in Computer Science, S. Agrawal and D. Lin (eds.), pp. 188-215, Springer-Verlag, 2022.

I. Iliashenko, M. Izabachène, A. Mertens and H. V. L. Pereira,
Homomorphically counting elements with the same property,
Proceedings on Privacy Enhancing Technologies (PETS), Volume 2022 (2022): Issue 4, pp. 670-683, PoPETS, 2022

H. Chen, I. Iliashenko and K. Laine,
When HEAAN Meets FV: a New Somewhat Homomorphic Encryption with Reduced Memory Overhead,
Proceedings of the 18th IMA International Conference on Cryptography and Coding (IMA CC), pp. 265-285, Springer-Verlag, 2021

I. Iliashenko, C. Nègre and V. Zucca,
Integer Functions Suitable for Homomorphic Encryption over Finite Fields,
Proceedings of the 9th on Workshop on Encrypted Computing & Applied Homomorphic Cryptography (WAHC), pp. 1-10, ACM, 2021

K. Cong, R. Cruz Moreno, M. B. da Gama, W. Dai,
I. Iliashenko, K. Laine and M. Rosenberg,
Labeled PSI from Homomorphic Encryption with Reduced Computation and Communication,
Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security (ACM CCS), pp. 1135-1150, ACM, 2021.

I. Iliashenko and V. Zucca,
Faster Homomorphic Comparison Operations for BGV and BFV,
Proceedings on Privacy Enhancing Technologies (PETS), Volume 2021 (2021): Issue 3 (July 2021), pp. 246-264, Sciencdo, 2021.

C. Bonte and I. Iliashenko,
Homomorphic String Search with Constant Multiplicative Depth,
 Proceedings of the 2020 ACM SIGSAC Conference on Cloud Computing Security Workshop (CCSW), pp. 105-117, ACM, 2020.

C. Bootland, W. Castryck, I. Iliashenko and F. Vercauteren,
Efficiently Processing Complex-Valued Data in Homomorphic Encryption,
 Journal of Mathematical Cryptology 14 (1, Special Issue Mathcrypt 2018): 55-65, 2020.

W. Castryck, I. Iliashenko and F. Vercauteren,
Homomorphic SIM2D Operations: Single Instruction Much More Data,
 Advances in Cryptology - EUROCRYPT 2018 - 37th Annual International Conference on the Theory and Applications of Cryptographic Techniques, Tel Aviv, Israel, April 29 - May 3, 2018 Proceedings, Part I, volume 10820 of Lecture Notes in Computer Science, J. Nielsen and V. Rijmen (eds.), pp. 338-359, Springer-Verlag, 2018.

C. Bonte, C. Bootland, J. W. Bos, W. Castryck, I. Iliashenko, and F. Vercauteren,
Faster Homomorphic Function Evaluation Using Non-Integral Base Encoding,
 In Cryptographic Hardware and Embedded Systems – CHES 2017 – 19th International Conference, Taipei, Taiwan, September 25-28, 2017, Proceedings, volume 10529 of Lecture Notes in Computer Science, W. Fischer, and Naofumi Homma (eds.), pp. 579-600, Springer-Verlag, 2017.

J. W. Bos, W. Castryck, I. Iliashenko, and F. Vercauteren,
Privacy-friendly Forecasting for the Smart Grid Using Homomorphic Encryption and the Group Method of Data Handling,
 In Progress in Cryptology - AFRICACRYPT 2017 - 9th International Conference on Cryptology in Africa, Dakar, Senegal, May 24-26, 2017, Proceedings, volume 10239 of Lecture Notes in Computer Science, M. Joye, and A. Nitaj (eds.), pp. 184-201, Springer-Verlag, 2017.

W. Castryck, I. Iliashenko, and F. Vercauteren,
On Error Distributions in Ring-based LWE,
 LMS Journal of Computation and Mathematics 19 (Special Issue ANTS-XII), pp. 130-145, 2016.

W. Castryck, I. Iliashenko, and F. Vercauteren,
Provably Weak Instances of Ring-LWE Revisited,
 In Advances in Cryptology – EUROCRYPT 2016 – 35th Annual International Conference on the Theory and Applications of Cryptographic Techniques, Vienna, Austria, May 8-12, 2016, Proceedings, Part I, volume 9665 of Lecture Notes in Computer Science, J. Coron, and M. Fischlin (eds.), pp. 147-167, Springer-Verlag, 2016.

Talks/Demos

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| Dec. 2021 | When HEAAN Meets FV: a New Somewhat Homomorphic Encryption with Reduced Memory Overhead
IMA CC 2021 |
| Oct. 2021 | Private set intersection
via somewhat homomorphic encryption
FHE.org meetup |
| Aug. 2021 | Private set intersection
via somewhat homomorphic encryption
IKBFU, Kaliningrad, Russia |

Jul. 2021	Faster homomorphic comparison operations for BGV and BFV PETS 2021
Jun. 2020	Lattices in cryptography ANTS-XIV summer school
Dec. 2019	On error distributions in ring-based LWE IKBFU, Kaliningrad, Russia
Nov. 2019	Sparse-secret Ring-LWE in FHE: Is It Really Needed? London Lattice meeting Royal Holloway University, Egham, UK
Aug. 2019	Noise-free FHE Crypto Lunch meeting Microsoft Research, Redmond, WA, USA
Aug. 2018	Efficiently processing complex-valued data in homomorphic encryption Mathcrypt 2018 Santa Barbara, CA, USA
May 2018	Secure smart meter demo Imec Technology Forum Antwerp, Belgium
May 2018	Secure smart meter demo HEAT project final review meeting Leuven, Belgium
Jul. 2018	w-NIBNAF for faster evaluation in SHE schemes IKBFU, Kaliningrad, Russia
May 2017	Privacy-friendly forecasting for the smart grid using homomorphic encryption AFRICACRYPT 2018 Dakar, Senegal
Aug. 2016	On error distributions in ring-based LWE ANTS-XII Kaiserslautern, Germany
Nov. 2016	Privacy-friendly forecasting for the smart grid using homomorphic encryption Colloquium Coding Theory and Cryptography Brussels, Belgium
Research stays	
Nov. 2019	Information Security Group, Royal Holloway University, UK Topic: noise analysis of FHE schemes
Jun. 2019 – Sep. 2019, Jun. 2018 – Sep. 2018	Cryptography and Privacy Research group, Microsoft Research, Redmond, WA, USA Topic: optimization and implementation of the HEAAN HE scheme in the SEAL library
Feb. 2012	Institute of Computer Science, University of Leipzig, Germany Topic: cryptography based on AG-codes

	Oct. 2011 – Jan. 2012	Institute of Mathematics and Computer Science, University of Greifswald, Germany Topic: applied mathematics
Teaching	Spring 2019, Spring 2018	Advanced Crypto Teaching assistant Practice session on quantum algorithms KU Leuven, Belgium
	Fall 2014	Geometric codes Lecturer IKBFU, Kaliningrad, Russia
Students	2017 – 2018	Robbe Motmans Master of Science in Mathematics “Analysis and simulations of Shor’s algorithm” Department of Mathematics, KU Leuven, Belgium
	2020-2021	Pieterjan Thijs Master of Science in Mathematics “Conversion algorithms between homomorphic encryption schemes” Department of Mathematics, KU Leuven, Belgium Jiayi Kang Master of Science in Mathematics “Efficient Homomorphic Encryption for Fixed Point Arithmetic” Department of Mathematics, KU Leuven, Belgium Helena Heerwegh Master of Science in Mathematics “Groups of Unknown Order” Department of Mathematics, KU Leuven, Belgium Wannes Manhaeve Master of Science in Artificial Intelligence “Training least squares support vector machines with homomorphic encryption” Department of Electrical Engineering, KU Leuven, Belgium
Grants	Oct. 2019 – Aug. 2021	FWO junior postdoctoral fellowship Project: Analysis of privacy-friendly pattern matching using homomorphic encryption
	Feb. 2012	DAAD - Leonhard Euler Scholarship
Seminars	Aug. 2015 – present	COSIC seminar Public-key group meeting Computation on Encrypted Data (CoED) meeting ESAT, KU Leuven, Belgium
	Jun. 2019 – Aug. 2019 Jun. 2018 – Aug. 2018	Crypto Lunch meetings Cryptography and Privacy Research group, Microsoft Research, Redmond, USA
Reviews	Conferences	CHES 2016 Asiacrypt 2016, 2017, 2019, 2021, 2022 SAC 2016

References

	ArcticCrypt 2016 Eurocrypt 2017-2021 Crypto 2017, 2018, 2020, 2021 PKC 2018-2020, 2022 ACNS 2018, 2020 CT-RSA 2020, 2021 USENIX 2021
Workshops	Waifi 2016 WIFS 2017 WAHC 2022 (PC member) WAHC 2024 (PC member)
Journals	International Journal of Information Security Journal of Cryptology IEEE Transactions on Information Forensics and Security Designs, Codes and Cryptography
Prof. Frederik Vercauteren	ESAT/COSIC, KU Leuven Kasteelpark Arenberg 10 3001, Leuven, Belgium +32 16 37 60 80 frederik.vercauteren@kuleuven.be
Dr. Kim Laine	Microsoft Research 14820 NE 36th Street, Building 99 98052, Redmond, Washington, USA kim.laine@microsoft.com
Dr. Ilya Razenshteyn	Ciphermode Labs Inc. 4470 W Sunset Blvd Suite 107 PMB 92370 90027, Los Angeles, California, USA ilya.razenshteyn@ciphermode.tech
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